MacEwan UNIVERSITY

CMPT 315

Lab 6 – Asynchronous Coding, Browser APIs & External APIs

General Information

- **Submission due date**: Due **@10:59 AM on 27-Oct-2026** (right before your next lab)
- **Lab weight**: 2%

OVERVIEW:

In this lab, you will build a **trivia game** that fetches multiple-choice questions from Open Trivia DB using async/await. The player selects a difficulty level (easy/medium/hard), completes one round, then sees a final score and an answer review. The best score is saved in localStorage.

This lab practices the following JavaScript concepts:

- DOM manipulation:
  - document.getElementById()
  - reading .value
  - updating .textContent
- Event handling:
  - addEventListener('click', ...) to trigger actions
- Async JavaScript:
  - async/await with fetch() for API calls
- JSON + persistence:
  - JSON.stringify / JSON.parse with localStorage

PART 1: Setting Up

Complete these steps before you start coding.

1. Create a folder: trivia-game
2. In VS Code: File -> Open Folder -> select trivia-game
3. Import files (from mêskanâs): **index.html**, **style.css** and **script.js**

PART 2: Your Tasks

You are provided with the starter files. Follow the TASK comments in them.

Task A — Form Validation

- In validateStartForm(), clear previous error messages in errName and errEmail.
- Validate Name: must be 2–30 characters after trim().
- Validate Email: must follow a basic valid shape (example: a@b.com).
- Display errors on the page (do not use alert for validation).
- Return true if valid; otherwise return false.

Task B — Save/Load Best Score

- loadProgress(): read localStorage using STORAGE_KEY. If missing or invalid JSON, return { bestScore: 0, bestTotal: 0, attempts: 0 }.
- saveProgress(obj): store the object under STORAGE_KEY using JSON.stringify().

Task C — External API Integration

- getToken(): call fetch(API.token) using await. Throw an error if the HTTP response is not OK or the token is missing.
- fetchQuestions(cfg): build the URL using API.questionsUrl(...). Fetch JSON and handle response_code: 0 (success) and 5 (rate limit).

Task D — Difficulty Selection

- Use the Difficulty dropdown (id="mode") which provides: easy, medium, hard.
- Before fetching questions, set the round difficulty based on the selection (ROUND.difficulty).
- The quiz title should reflect the selected difficulty (the starter already displays ROUND.difficulty in the title).

Task E — Show Answer Review

- Save the player's selected answer for each question (store it on the question object).
- On the Result screen, show an answer review listing each question, the player's answer, the correct answer, and ✓/✗.
- The starter provides a partial review builder in finishRound(). Complete the missing TODO lines.

Task F — Creative UX & Design

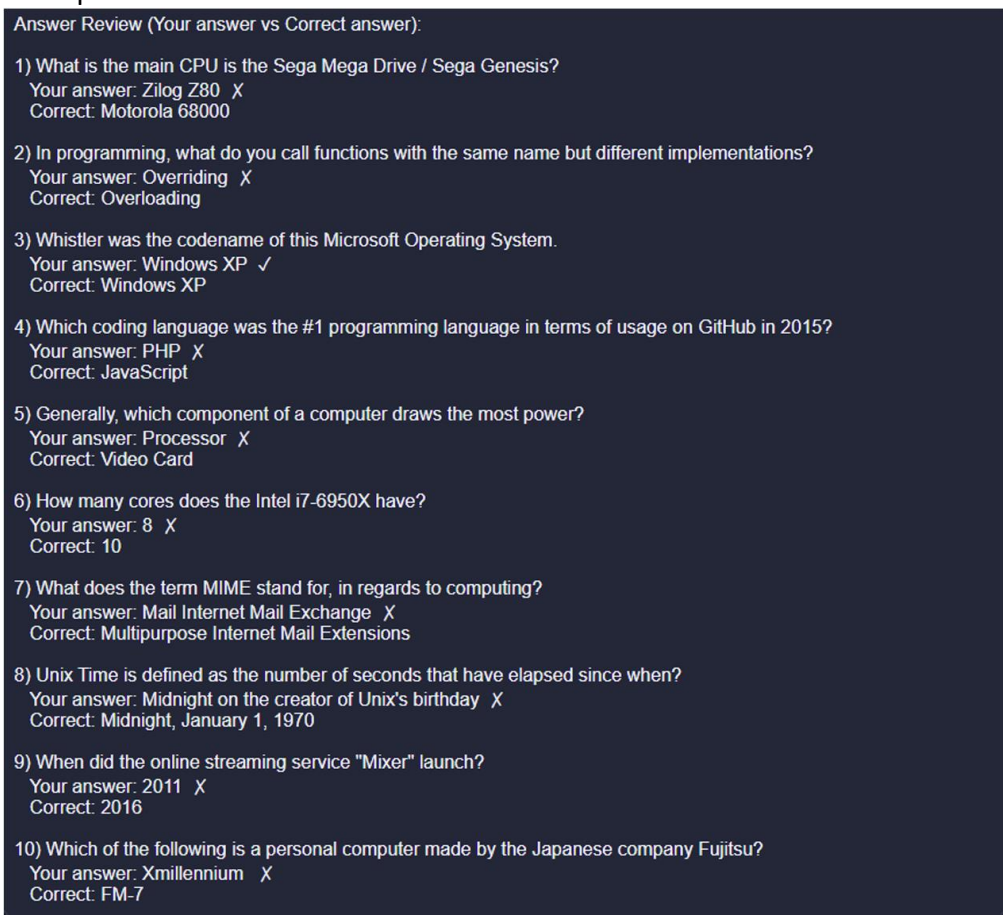
- Rename the site: update the <h1> to a unique app name.
- Logo/banner: add an image or a CSS-styled element in the header area.
- Theming (optional): you can customize colors, fonts, and button styling in the provided CSS.

PART 3: Local Testing + Deployment

Before you deploy your website on GitHub, you must first confirm it works locally. Once local testing succeeds you will go ahead and deploy. You can follow these steps:

1. **Save your files**
    - Windows: **Ctrl + S**
    - Mac: **Cmd + S**

2. **Test Option A (Highly Recommended): Use Live Server in VS Code**
    - Right-click index.html → **Open with Live Server**
    - Your browser will open with a local address like http://127.0.0.1:5500/
    - Save changes and the page updates automatically (or refresh if needed)
3. **Test Option B (Simple): Open the file directly**
    - In VS Code, right-click index.html
    - Choose **Reveal in File Explorer** (Windows) / **Reveal in Finder** (Mac)
    - Double-click index.html to open it in your browser
4. **Focus/Blur Checklist**
    - Invalid name/email shows on-page errors (no alerts for validation).
    - Difficulty selection affects which questions are fetched (easy/medium/hard).
    - Click Start Quiz → Loading appears → first question appears.
    - After answering all questions → Result screen shows score X/N.
    - Answer review shows each question, your answer, correct answer, and ✓/✗. Here is a sample:



    - Best score is saved and still visible after a page refresh.
    - Reset Best Score clears saved values in this browser.
5. **Deploy to the Internet**
    - Refer to previous labs or use your preferred steps to **go live** on GitHub Pages.

SUBMISSION

Submit **one PDF document** through mêskanâs containing your answers to the questions below:

1. Paste your **GitHub Repo URL** (e.g., https://github.com/username/trivia-game)
2. Paste your live **GitHub Pages URL** (e.g., https://username.github.io/trivia-game)
3. Why do we need **async/await** when using **fetch()**?
4. What does OpenTDB **response_code** mean in your implementation? Explain 0 and 5 mean.
5. What exactly is stored under **STORAGE_KEY** and why is **JSON** used?

DEMO: Tuesday, October 27, 2026 @11:00 AM

- Demonstrate your completed lab during the lab session.
- Be prepared to explain your solution and answer questions about your code.

MARKING SCHEME

Component	Mark
Lab Questions (PDF Submission)	5
Lab Demo	15
Total	**20**